



**MINISTERE DE L'ENSEIGNEMENT SUPERIEUR ET DE LA
FORMATION PROFESSIONNELLE**

**INSTITUT DE FORMATION ET DE RECHERCHE EN
INFORMATIQUE**

.....
Filière : Sécurité Informatique

**TP de DNS-Configuration d »une zone
avec serveur DNS et serveur Web
distincts**

Réalisé par :

GANSE Gille-Christ

Sous la supervision de :

Mr FAGLA Natus

ANNÉE ACADÉMIQUE : 2025 – 2026

PLAN

Présentation des informations (adresses) de la machine

I. Déclaration des zones dans Bind

II. Création des fichiers de zone

Zone directe

Zone inverse

III. Tests dig

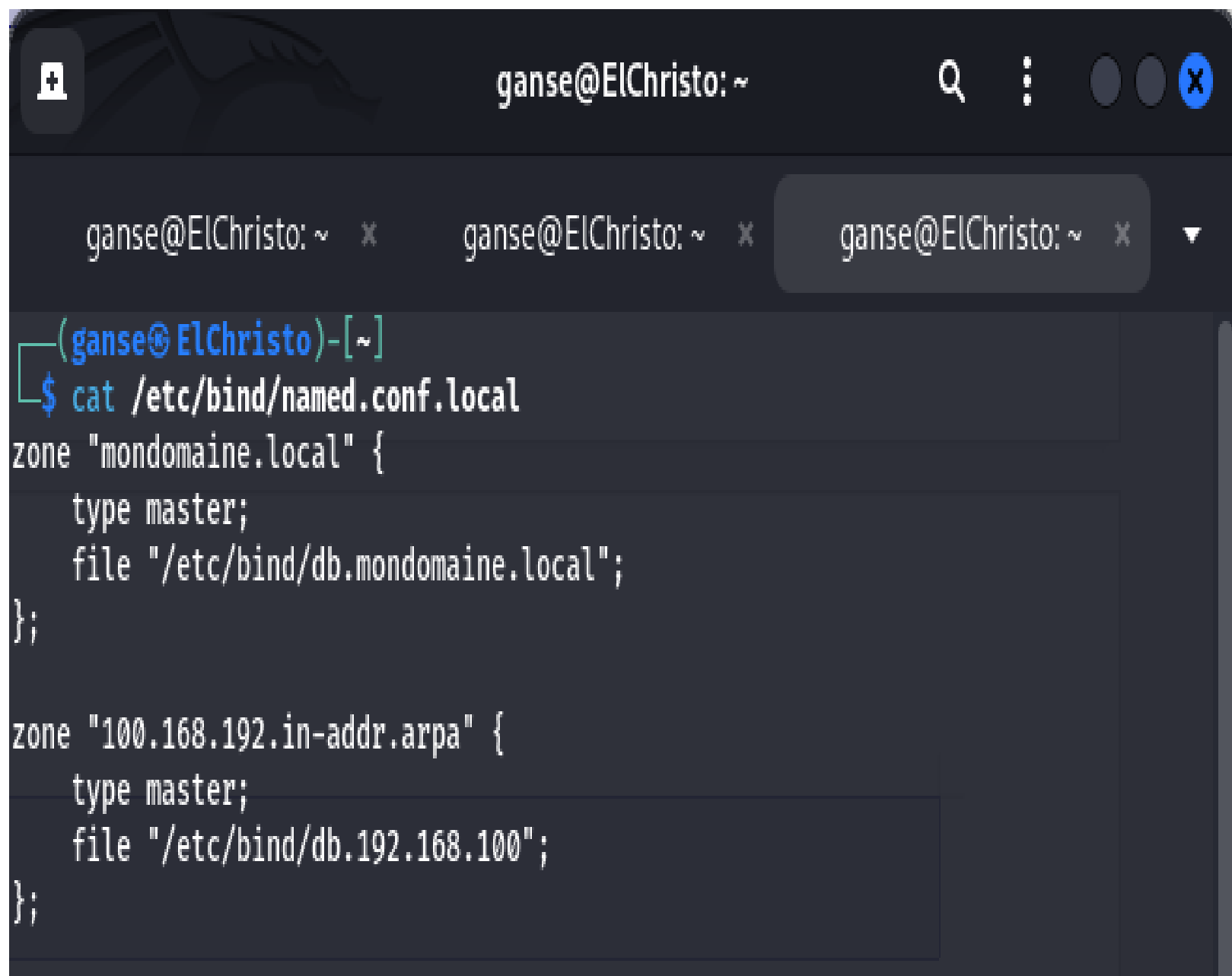
Zone direct

Zone inverse

Présentation des informations (adresses) de la machine

```
ganse@ElChristo: ~  
ganse@ElChristo: ~ x ganse@ElChristo: ~ x ganse@ElChristo: ~ x  
_default/100.168.192.1n-addr.arpa/IN: unknown class/type  
  
(ganse@ElChristo)-[~]  
$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc fq_codel state DOWN group default qlen 1000  
    link/ether 10:62:e5:4c:8b:fe brd ff:ff:ff:ff:ff:ff  
3: wlan0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000  
    link/ether 7c:76:35:d8:22:91 brd ff:ff:ff:ff:ff:ff  
    inet 192.168.100.6/24 brd 192.168.100.255 scope global dynamic noprefixroute wlan0  
        valid_lft 85451sec preferred_lft 85451sec  
    inet6 fe80::7e76:35ff:fed8:2291/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever
```

Déclaration des zones dans Bind

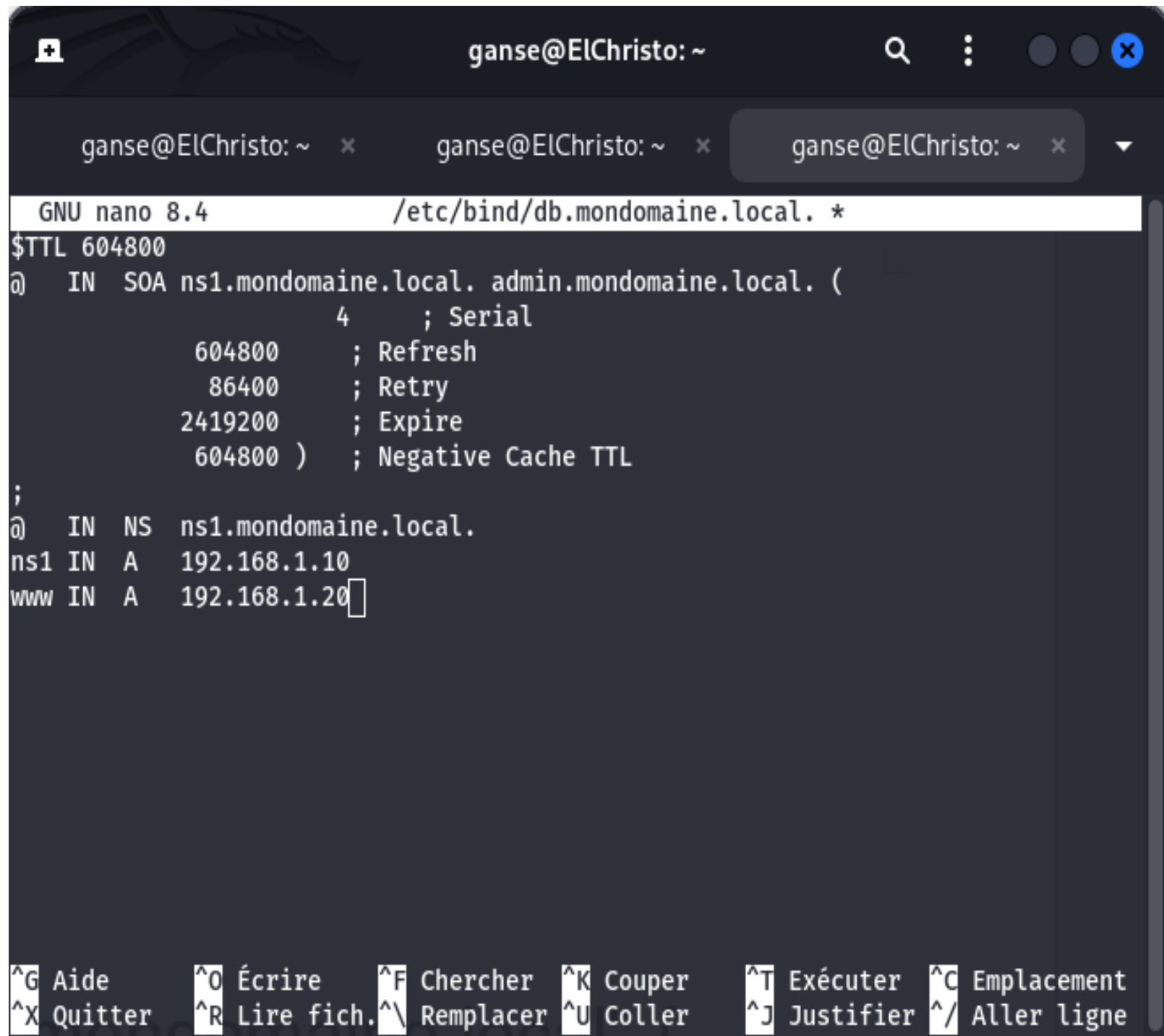


A terminal window titled "ganse@ElChristo: ~" with a search icon, a menu icon, and window control buttons. It shows three tabs, all with the same title. The active tab displays the output of the command `cat /etc/bind/named.conf.local`. The output shows two zone declarations in a BIND configuration file.

```
(ganse@ElChristo)-[~]  
$ cat /etc/bind/named.conf.local  
zone "mondomaine.local" {  
    type master;  
    file "/etc/bind/db.mondomaine.local";  
};  
  
zone "100.168.192.in-addr.arpa" {  
    type master;  
    file "/etc/bind/db.192.168.100";  
};
```

Création des fichiers de zone

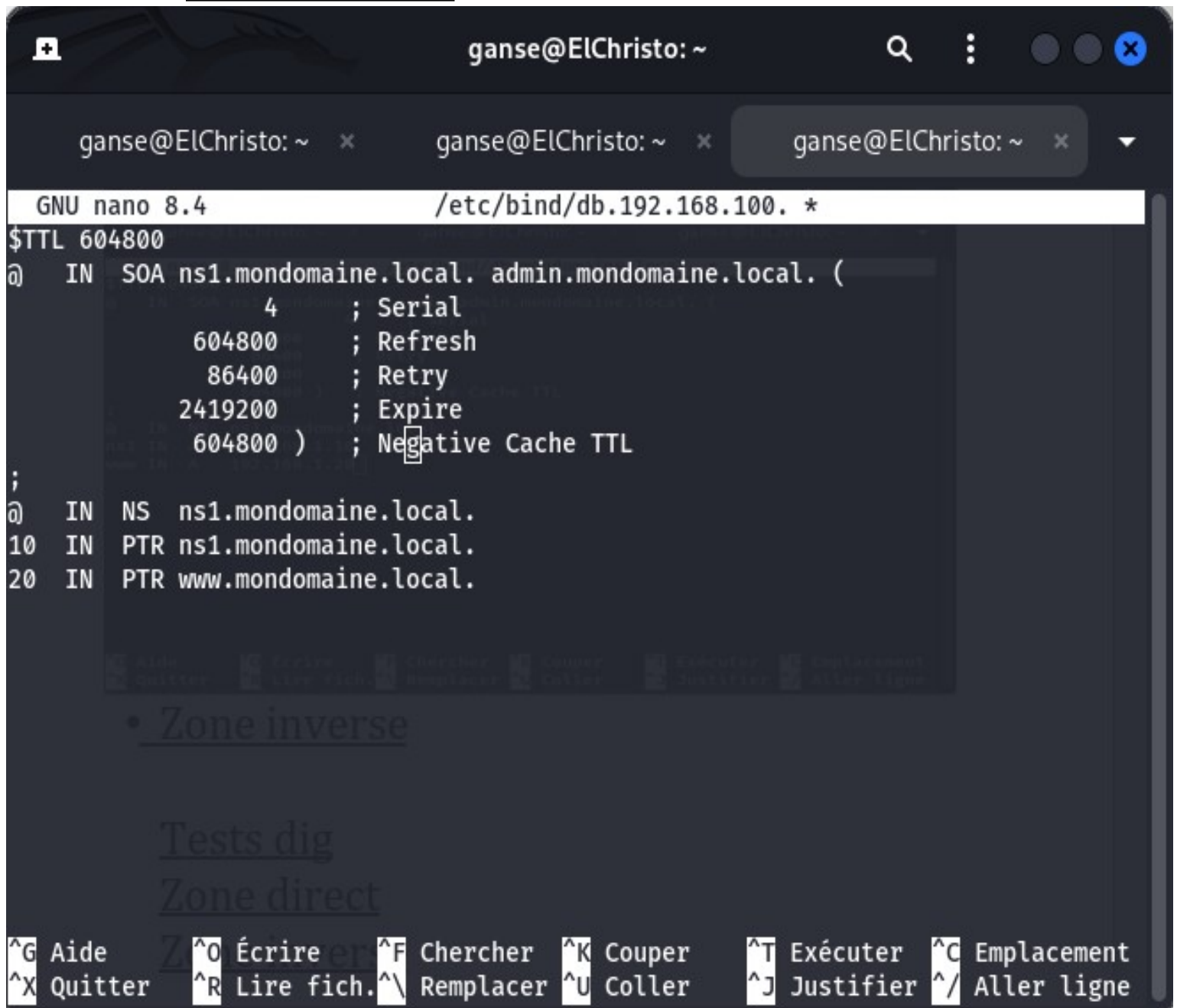
- Zone directe



The screenshot shows a terminal window with a dark theme. The title bar indicates the user is 'ganse@ElChristo' in the home directory. The terminal is running 'GNU nano 8.4' and editing the file '/etc/bind/db.mondomaine.local.*'. The content of the file is a BIND zone file for 'mondomaine.local'. It starts with a \$TTL of 604800, followed by an SOA record for 'ns1.mondomaine.local.' with an administrator email of 'admin.mondomaine.local.'. The SOA record includes a serial number of 4, a refresh interval of 604800, a retry interval of 86400, an expire time of 2419200, and a negative cache TTL of 604800. Below the SOA record, there is an NS record for 'ns1.mondomaine.local.' and two A records: 'ns1' with IP '192.168.1.10' and 'www' with IP '192.168.1.20'. The cursor is at the end of the 'www' A record. At the bottom of the terminal, there is a help bar with various keyboard shortcuts for nano editor functions.

```
ganse@ElChristo: ~
ganse@ElChristo: ~
ganse@ElChristo: ~
GNU nano 8.4 /etc/bind/db.mondomaine.local.*
$TTL 604800
@ IN SOA ns1.mondomaine.local. admin.mondomaine.local. (
                        4      ; Serial
                        604800 ; Refresh
                        86400  ; Retry
                        2419200 ; Expire
                        604800 ) ; Negative Cache TTL
;
@ IN NS ns1.mondomaine.local.
ns1 IN A 192.168.1.10
www IN A 192.168.1.20
^G Aide      ^O Écrire   ^F Chercher ^K Couper   ^T Exécuter ^C Emplacement
^X Quitter  ^R Lire fich.^\\ Remplacer ^U Coller   ^J Justifier ^/ Aller ligne
```

- Zone inverse



The screenshot shows a terminal window with a dark theme. At the top, the title bar reads 'ganse@ElChristo: ~'. Below it, the terminal window title is 'ganse@ElChristo: ~'. The main content is a nano editor window editing the file '/etc/bind/db.192.168.100.*'. The editor shows the configuration for an inverse DNS zone. The first line is '\$TTL 604800'. The second line is '@ IN SOA ns1.mondomaine.local. admin.mondomaine.local. ('. The third line is '4 ; Serial'. The fourth line is '604800 ; Refresh'. The fifth line is '86400 ; Retry'. The sixth line is '2419200 ; Expire'. The seventh line is '604800) ; Negative Cache TTL'. The eighth line is ';'. The ninth line is '@ IN NS ns1.mondomaine.local.'. The tenth line is '10 IN PTR ns1.mondomaine.local.'. The eleventh line is '20 IN PTR www.mondomaine.local.'. At the bottom of the terminal, there is a list of nano editor shortcuts: ^G Aide, ^X Quitter, ^O Écrire en, ^R Lire fich., ^F Chercher, ^\ Remplacer, ^K Couper, ^U Coller, ^T Exécuter, ^J Justifier, ^C Emplacement, ^/ Aller ligne.

```
ganse@ElChristo: ~
GNU nano 8.4 /etc/bind/db.192.168.100.*
$TTL 604800
@ IN SOA ns1.mondomaine.local. admin.mondomaine.local. (
    4 ; Serial
    604800 ; Refresh
    86400 ; Retry
    2419200 ; Expire
    604800 ) ; Negative Cache TTL
;
@ IN NS ns1.mondomaine.local.
10 IN PTR ns1.mondomaine.local.
20 IN PTR www.mondomaine.local.

^G Aide ^O Écrire en ^F Chercher ^K Couper ^T Exécuter ^C Emplacement
^X Quitter ^R Lire fich. ^\ Remplacer ^U Coller ^J Justifier ^/ Aller ligne
```

Tests dig

```
ganse@ElChristo: ~  
ganse@ElChristo: ~ x ganse@ElChristo: ~ x ganse@ElChristo: ~ x  
(ganse@ElChristo)-[~]  
$ dig www.mondomaine.local @127.0.0.1  
  
; <<>> DiG 9.20.18-1-Debian <<>> www.mondomaine.local @127.0.0.1  
;; global options: +cmd  
;; Got answer:  
;; WARNING: .local is reserved for Multicast DNS  
;; You are currently testing what happens when an mDNS query is leaked to DNS  
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 17789  
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1  
  
;; OPT PSEUDOSECTION:  
; EDNS: version: 0, flags:; udp: 1232  
; COOKIE: 343009948603cca7010000006991f153cca4b67c7fa9847f (good)  
;; QUESTION SECTION:  
;www.mondomaine.local. IN A  
  
;; ANSWER SECTION:  
www.mondomaine.local. 604800 IN A 192.168.1.7  
  
;; Query time: 4 msec  
;; SERVER: 127.0.0.1#53(127.0.0.1) (UDP)  
;; WHEN: Sun Feb 15 17:16:19 CET 2026  
;; MSG SIZE rcvd: 93
```

```
ganse@ElChristo: ~  
ganse@ElChristo: ~ x ganse@ElChristo: ~ x ganse@ElChristo: ~ x  
(ganse@ElChristo)-[~]  
$ dig -x 192.168.100.6 @127.0.0.1  
; <<>> DiG 9.20.18-1-Debian <<>> -x 192.168.100.6 @127.0.0.1  
;; global options: +cmd  
;; Got answer:  
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 62867  
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1  
;; OPT PSEUDOSECTION:  
;; EDNS: version: 0, flags:; udp: 1232  
;; COOKIE: e4c9bbbec929e33a010000006991f2a51a6086bd0efb85a7 (good)  
;; QUESTION SECTION:  
;6.100.168.192.in-addr.arpa. IN PTR  
;; ANSWER SECTION:  
6.100.168.192.in-addr.arpa. 604800 IN PTR ns1.mondomaine.local.  
;; Query time: 0 msec  
;; SERVER: 127.0.0.1#53(127.0.0.1) (UDP)  
;; WHEN: Sun Feb 15 17:21:57 CET 2026  
;; MSG SIZE rcvd: 117
```

```
ganse@ElChristo: ~  
ganse@ElChristo: ~ x ganse@ElChristo: ~ x ganse@ElChristo: ~ x  
(ganse@ElChristo)-[~]  
$ dig -x 192.168.100.7 @127.0.0.1  
;  
<<>> DiG 9.20.18-1-Debian <<>> -x 192.168.100.7 @127.0.0.1  
;; global options: +cmd  
;; Got answer:  
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 60601  
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1  
;  
;; OPT PSEUDOSECTION:  
; EDNS: version: 0, flags:; udp: 1232  
; COOKIE: 9f5303597b5d2ecb010000006991f30c071ac349f0d85dd3 (good)  
;; QUESTION SECTION:  
; 7.100.168.192.in-addr.arpa. IN PTR  
;  
;; ANSWER SECTION:  
7.100.168.192.in-addr.arpa. 604800 IN PTR www.mondomaine.local.  
;  
;; Query time: 4 msec  
;; SERVER: 127.0.0.1#53(127.0.0.1) (UDP)  
;; WHEN: Sun Feb 15 17:23:40 CET 2026  
;; MSG SIZE rcvd: 117
```